

FERMI TUTORIAL



Kerberos Authentication: A Practical SSH Guide

Jose Berlioz
jberlioz@fnal.gov

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1 Kerberos Crash Course

Kerberos centralizes authentication in the **Key Distribution Center (KDC)**. Users (principals) obtain a **Ticket Granting Ticket (TGT)** via `kinit`; services (like SSH) are accessed using **service tickets** issued by the Ticket Granting Service (TGS). Hosts decrypt tickets using a synced `/etc/krb5.keytab`.

Suggested References

- [MIT Kerberos](#): papers and user/developer guides.
- [Taming Kerberos](#) (Computerphile) video overview.
- Wikipedia: Kerberos overview.
- Fermilab: [ServiceNow: Encryption @ Fermilab](#).
- Beams-doc-1502 (Java-oriented, concepts still useful).

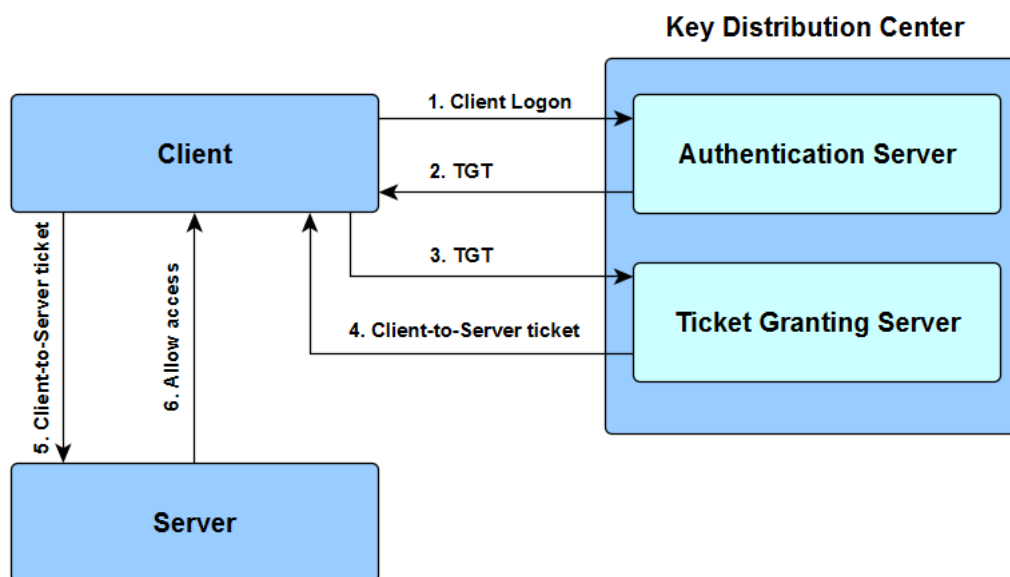


Figure 1: Kerberos flow: client gets TGT, requests a service ticket, then authenticates to `sshd` via GSSAPI.

1.1 Authentication

1.2 Principal (Client) Authentication

In Kerberos, client authentication is how a user proves their identity to the system without sending a password. The client gets a ticket from the Key Distribution Center

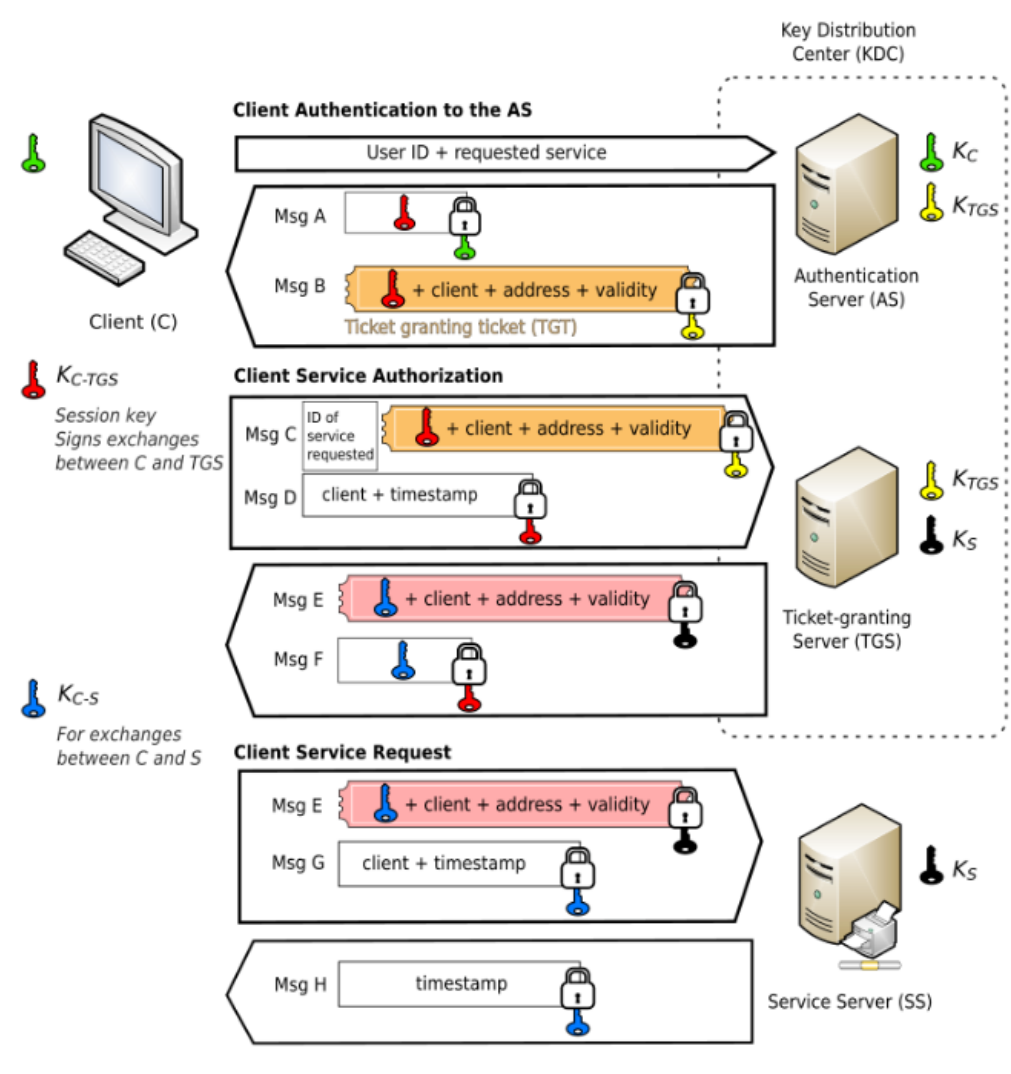


Figure 2: Kerberos negotiation. The keys are: K_C (client secret from password), K_{TGS} (TGS secret with AS), K_{C-TGS} (client-TGS session key), K_S (service server secret with TGS), and K_{C-S} (client-service session key).

(KDC) using a secret derived from the password. That ticket is then presented to services to prove identity, cryptographically and time-limited.

Since the ticket is used to get other tickets without reauthenticating again, its called Ticket-Granting Ticket (TGT).

```
kinit <username> #Command to request client ticket.
                  #Bc its used to get other tickets, its called
                  Ticket Granting Ticket (TGT)
klist            # verify TGT under default cache
```

1.3 Service Authentication

Let's assume a client wants to ssh into a host. In the client-service authentication, the client uses its TGT to request a service ticket from the KDC. The KDC verifies the TGT and issues a ticket encrypted for the target service. The client presents this service ticket to the service. The service validates it and grants access, with no password exchange.

```
ssh <hostname>.fnal.gov #Service Authentication happens
                        #automatically for ssh

klist                  # service ticket for
                        # host/<hostname>.fnal.gov now present
```

For that to work, the service must be pre-registered with the KDC and share a long-term secret key with it. The KDC uses that key to encrypt service tickets meant for the service and give them to the client.

Section 4.1. explains the process of registering with the KDC and setting up the files in the right place at Fermilab.

2 Goals

This guide explains how to configure SSH with Kerberos authentication on Linux systems (**tested on Ubuntu/Xubuntu 22.04**).

Section 3 is for setting up your kerberos credentials to log in into devices while Section 4 shows you how to setup the ssh environment so you can ssh to the devices.

3 Setup your client

Requesting Kerberos Client Access

1. Open the ServiceNow portal

Navigate to <https://fermi.servicenowservices.com/wp> and log in with your FNAL credentials.

2. Locate the Kerberos access request

Use the search bar to find “**Access to Kerberized Machines**” and open the corresponding request form.

3. Fill in the request form

Complete the required personal and system information fields.

4. Do not request principals

Ensure that **NONE** of the following options are selected:

- Host Principals
- FTP Principals
- HTTP Principals
- Kerberos Principal Instance for Root Access

This are used when getting your host/service or for getting access to root.

5. Add justification

In the **Additional Information** field, enter:

Need Kerberos client credentials for SSH

6. Submit and wait for approval

Submit the request and wait for the Service Desk to process it. Once approved, you will be able to obtain a Ticket-Granting Ticket (TGT) using `kinit` and authenticate to kerberized hosts via SSH.

4 Setting up your host

4.1 Kerberos Environment and Client Sanity Checks

Before configuring a Kerberos server, first make sure your environment and client setup are working:

- **Environment readiness:** Kerberos requires accurate time (tolerates at most 5 minutes of skew) and proper network access. Verify your host/client clock is synchronized and firewall rules permit inbound SSH from trusted subnets or IP ranges. Inside the control's network, the available ntp server is called `chablis.fnal.gov`. If that doesn't work, ask your network admin for the ntp server.
- **Client-side Kerberos check:** On a known kerberized host:

1. Run `kinit <username>` and enter your Kerberos password.
2. Confirm a valid Ticket Granting Ticket (TGT) exists with:

```
klist
```

3. Test SSH to another Kerberos-enabled host:

```
ssh <hostname> .fnal.gov
```

A successful login without a password prompt confirms Kerberos is working.

- **Optional (AlmaLinux utilities):**

- Documentation and package notes: <https://linux-mirrors.fnal.gov/linux/fermilab/el/9/notes.html>
- Helpful RPMs: `fermilab-conf_kerberos`, `fermilab-util_makehostkeys`

Only after these checks succeed should you proceed with server-side setup and troubleshooting.

4.2 Before Connect Host to Internet

4.2.1 Request Network Information

- Submit a network hookup request via: <https://fermi.servicenowservices.com>.
 - Be sure to request: IP address, netmask, gateway, and DNS servers (these are not always provided unless explicitly asked for).
- Sometimes the Service Desk reply only includes the IP address, and it is easy to lose track of the hostname you originally requested. To recover it, run:

```
nslookup <assigned-ip>
# Example:
nslookup 131.225.113.200
# ... name = test-system.fnal.gov.
```

- If you did not receive the IP address at all, follow up with the Service Desk by email to request it.
- Use only lowercase letters in hostnames—Kerberos is case sensitive.
- To search for registered hostnames, see: <https://nimi.fnal.gov/nimi/>.

4.2.2 Set Hostname Files

```
sudoedit /etc/hostname
# set to: test-system
```

```
sudoedit /etc/hosts
# ensure:
# 127.0.0.1          loopback
# 131.225.113.200    test-system.fnal.gov  test-system
```

4.2.3 Configure Network (Example: using Netplan on Ubuntu 22.04)

```
sudoedit /etc/netplan/01-netcfg.yaml
# Example static config (adjust for your env):
# network:
#   version: 2
#   ethernets:
#     enp1s0:
#       addresses: [131.225.113.200/24]
#       routes:
#         - to: default
#           via: 131.225.113.1
#       nameservers:
#         search: [fnal.gov]
#         addresses: [131.225.8.120, 131.225.8.121]
sudo netplan apply
```

We will use this hostname later when requesting the host principal. If your system uses

4.2.4 OpenSSH Daemon (sshd) Configuration

Even before starting testing Kberberos, edit the SSH configuration. Fermilab does not allow SSH Public Key authentication and SSH Password Authentication. Instead, Kerberos-related services should be enabled. Edit `/etc/ssh/sshd_config` to include:

```
Protocol 2
RSAAuthentication no
PubkeyAuthentication no
PasswordAuthentication no
ChallengeResponseAuthentication no
UsePAM yes
KerberosAuthentication yes
KerberosOrLocalPasswd no
KerberosTicketCleanup yes
GSSAPIAuthentication yes
GSSAPIKeyExchange yes
GSSAPICleanupCredentials yes
X11Forwarding yes
```

Restart SSH after changes:

```
sudo systemctl restart ssh
```

4.3 Setup after Network Connection

4.3.1 Install Required Packages (Ubuntu)


```
sudo apt-get update
sudo apt-get install krb5-config krb5-user krb5-doc
sudo apt-get install openssh-server
```

If packages differ, find the equivalent `krb5` and `openssh-server` for your distro.

4.3.2 Kerberos Configuration (`krb5.conf`)

1. Replace your `krb5.conf` with the matching Fermi config.¹

- <https://authentication.fnal.gov/krb5conf/>

2. Test KDC visibility:

```
kinit johnsmith
klist      # expect user TGT if successful
```

4.4 Local Account, Host Keys, and `.k5login`

4.4.1 Create a Matching Local User

```
# username must match Kerberos principal's local part (e.g.,
  johnsmith)
sudo adduser johnsmith
```

4.4.2 Generate SSH Host Keys (if needed)

```
sudo ssh-keygen -A
# If asked for passphrases, leave blank to avoid auth conflicts.
```

4.4.3 Authorize Kerberos Principals via `.k5login`

The `.k5login` file defines which Kerberos principals are authorized to log in to the previously created **local account** on your host. Each line should contain one principal in the format `USER@FNAL.GOV`.

```
# Create or update ~/.k5login with your Kerberos principal
# Replace [user1] with your local username
echo "[user1]@FNAL.GOV" | sudo tee -a /home/[user1]/.k5login

# Set correct ownership and restrictive permissions
su - [user1]
```

¹On Ubuntu, using the Scientific Linux variant is acceptable.

```
sudo chmod 600 /home/[user1]/.k5login
sudo chown [user1]:[user1] /home/[user1]/.k5login

# (Optional) grant access to an additional principal
echo "[user2]@FNAL.GOV" | sudo tee -a /home/[user1]/.k5login
```

After running the above commands, your .k5login should look like this.

```
# FILE: .k5login
johnsmith@FNAL.GOV
jeffbezoz@FNAL.GOV
```

4.5 Request a Host Principal

At this step, we use a one-time password issued by the KDC for the host principal to generate a permanent service key. The `ktadd` command stores this key in `/etc/krb5.keytab`, allowing the host (e.g. `sshd`) to accept Kerberos tickets without requiring a password again.

On <https://servicedesk.fnal.gov>, submit “Access to Kerberized Machines” and request `host/<hostname.domain>` using your registered host info.

Password Generation Update

Passwords arrive via `no_reply_auth@fnal.gov`. Construct it as:

- Username: first two letters (lowercase).
- Day: three-letter abbreviation, capitalized first letter.
- Date: MMDD (two digits each).
- Year: last two digits.
- Seconds: two digits from request time.
- Special char by day: Mon=%, Tue=\$, Wed=#, Thu=@, Fri=!, Sat=/, Sun=?.

Example: johndoe, Thu, Jan 1, 2001, 00s → joThu01010100@.

If all else fails, contact the Service Desk.

4.6 Install Service Key (Keytab) on the Host

Run as root (or via `sudo`) so `/etc/krb5.keytab` is written. If permissions stop the `kadmin` from generating the keytab, one must request another password. Replace `<hostname.domain>` with your specific `test-system.fnal.gov`.

```
sudo su
kadmin -p host/<hostname.domain> \
  -q "ktadd host/<hostname.domain>"
```

Notes:

- Kerberos is **case sensitive**; DNS is not. Use the exact DNS hostname:

```
nslookup 131.225.113.200
# ... name = test-system.fnal.gov.
```

- You will be prompted for the password from the prior step; success creates/updates /etc/krb5.keytab.
- AD helper script: `fermilab-util_makehostkeys` (same mechanism with safeguards).

4.6.1 Authenticate Host with `krb5.keytab`

Once the host principal has been created and the keytab is installed under /etc/krb5.keytab, you can test Kerberos authentication by obtaining a ticket using the keytab.

```
# Verify the keytab contents
sudo klist -k /etc/krb5.keytab

# Authenticate using the host principal from the keytab
sudo kinit -k -t /etc/krb5.keytab host/yourhost.fnal.gov@FNAL.GOV

# Confirm that a valid ticket was obtained
sudo klist
```

Notes:

- Replace `host/yourhost.example.com@REALM.EXAMPLE` with the exact principal listed in the keytab (often `host/$(hostname -f)@FNAL.GOV`).
- Ensure the keytab is readable only by root (permissions 600).
- If an old ticket is present, you may clear it with `kdestroy` before re-running `kinit`.

The commands below implement automatic Kerberos authentication during the boot process using a systemd service.

```
# 1. Create systemd service
sudo vim /etc/systemd/system/krb5-host.service
```

```
# 2. Put the content below in the service file. Service content:
[Unit]
```

```
After=network.target
```

```
[Service]
```

```
Type=oneshot
```

```
ExecStart=/usr/bin/kinit -k -t /etc/krb5.keytab \  
    host/hostname.fnal.gov@FNAL.GOV
```

```
[Install]
```

```
WantedBy=multi-user.target
```

```
# 3. Enable and start service
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable krb5-host.service
```

```
sudo systemctl start krb5-host.service
```

```
# 4. Verify ticket
```

```
klist
```

At each reboot, systemd triggers the service, which executes kinit using the keytab to obtain a Kerberos ticket for the host principal.

4.7 Troubleshooting Tips

Client-side verbosity

```
ssh -v <hostname>.fnal.gov
```

Server-side logs

```
sudo journalctl -u ssh --since "1 hour ago"  
# or:  
sudo tail -f /var/log/auth.log
```

Common Errors (from verbose SSH)

Key version is not available KDC needs to update. Try again in 1 hour.

UNKNOWN SERVER Using IP or unregistered/incorrect hostname. Run nslookup and use the exact DNS name.

LOOKING UP SERVER KDC cannot match host principal to DNS (often case/format mismatch). Align DNS hostname and Kerberos principal; avoid capitals.

Connection closed by UNKNOWN port 65535 Check the following:

- firewall is not blocking the sshd connection
- kerberos host principal is validated
- the sshd service is turned on (`systemctl status ssh.service`)

4.8 Summary

- Set host identity:
 - Update `/etc/hostname`, `/etc/hosts`, and `netplan`
- Install **Kerberos** and **OpenSSH**
- Add Fermilab's `krb5.conf`
- Configure `sshd` for Kerberos/GSSAPI
- Ensure:
 - Matching local user account
 - Valid `.k5login` file
- Request `host/<hostname.domain>` principal and add with `ktadd` to `/etc/krb5.keytab`
- Test setup using:
 - `kinit`
 - `klist`
 - `ssh -v`
- Troubleshooting:
 - Verify system clock / time synchronization
 - Check DNS ↔ principal name consistency